Atharv Shinde

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.34
Vivekanand College, Kolhapur	Class XII (Maharashtra State Board of Secondary & Higher Secondary Education)	2020	91.08%
St. Xavier's Highschool, Kolhapur	Class X (Maharashtra State Board of Secondary & Higher Secondary Education)	2018	90.80%

WORK EXPERIENCE

Teaching Assistant | IIT Hyderabad

- Conduct lab sessions and support course delivery for Computer Architecture, Algorithms, and Advanced Data Structures & Algorithms, including grading, exam coordination, and doubt resolution.

Software Development Engineer Intern | Mastercard, Pune

- Worked on the Individual Data Access Portal, a globally deployed system handling Mastercard users' personal data.
- Developed automation test suites using Selenium, debugged network request failures, integrated RESTful API workflows for QA pipelines, and managed compliance documents.

SKILLS

Programming & Software: Python, C++, C, Data Structures & Algorithms, Object-Oriented Programming, REST APIs

ML Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, NumPy, Pandas

Deep Learning & NLP: CNNs, RNNs, LSTMs/GRUs, Transformers, Attention Mechanisms, Vision Transformers (ViT), Transfer Learning, Natural Language Inference, Tokenization, Embeddings

Advanced ML: Federated Learning, Non-IID Learning, Task Heterogeneity, Pseudo Labeling, Semi-Supervised Learning, Multi-Label Classification

Tools & Platforms: Matplotlib, Seaborn, NLTK, SLURM, Git, Docker, MATLAB, Proteus

RELEVANT COURSES

Algorithms & Systems: Advanced Data Structures & Algorithms, Computer Networks, Network Security

Mathematics: Linear Algebra, Probability, Calculus, Discrete Mathematics

Machine Learning: Foundations of Machine Learning, Deep Learning, Visual Computing, Representation Learning

PROJECTS

Federated Semi-Supervised Medical Image Captioning | M.Tech Thesis (Ongoing)

PyTorch, Hugging Face Transformers, BLIP-2

- Investigated pseudo label collapse in a federated, semi-supervised medical image captioning pipeline under non-IID

clinical clients, identifying systematic overprediction of high-frequency findings.
- Built on BLIP-2's Q-Former with selective parameter aggregation and confidence based pseudo labeling for unlabeled medical images.
- Designed a label grounding mechanism to filter generated captions against structured clinical findings before training on pseudo labels, reversing the collapse.
- Validated the failure and correction using two independent labelers, showing that standard NLG metrics could not distinguish the collapsed and corrected models and motivating evaluation at the category level.

AdaptFedMLP | Federated Multi-Label Classification under Task Heterogeneity

PyTorch, NumPy, scikit-learn

- Reproduced and extended FedMLP (Sun et al., MICCAI 2024) for multi-label intracranial hemorrhage classification under strict task heterogeneity across five federated clients.
- Diagnosed gradient collapse from pseudo labels in Stage 2 training and developed AdaptFedMLP with multi-window CT preprocessing, adaptive thresholds, prototype smoothing, decoupled heads, and confidence-weighted aggregation.
- Improved macro AUROC from 0.414 after Stage 1 to 0.865 and BACC from 0.420 to 0.781 after adaptation, approaching the paper benchmark within 2 points using 50 vs. 450 Stage 2 rounds.
- Extended the study with SimCLR pretraining of the image encoder on medical images, improving AUROC from 0.866 to 0.901, with gains across all five classes; compared with a matched compute denoising autoencoder control.

Secure P2P Payment Platform

PHP, MySQL, Docker, Apache

- Built and hardened a Dockerized peer-to-peer payment platform - user authentication, transfers, and profile management - with a five member team.
- Mitigated OWASP Top 10 vulnerabilities via parameterized queries, CSRF tokens, and race condition safe transaction handling to prevent double spending.
- Exploited vulnerabilities in peer teams' platforms during a live red-team exercise, while patching issues identified in their own system.